

Cole Bregman

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Education

Columbia University

New York, NY · 2023 – 2027

B.S. Mechanical Engineering, Minor in Computer Science | GPA: 4.0 | Dean's list | Tau Beta Pi

Work Experience

Apple Inc.

Cupertino, CA

Manufacturing Design Engineering Intern

May 2025 - Present

- Saved \$2M+ to date through DFM optimizations and targeted design changes using Siemens NX, increasing yield.
- Enabled critical proto build by resolving 0% yield assembly blocker in 10 days through systematic root-cause analysis, cross-functional coordination with design and manufacturing, and on-site factory support; delivered on schedule.
- Achieved 4x improvement in critical strength metric by designing systematic DOE experiments, analyzing data with JMP and Python, and collaborating with Chinese vendors to remedy previously failing injection-molded components.
- Communicated new US manufacturing partnership to Apple CEO, factory leadership, and press by designing and building hands-on tabletop factory demonstration that effectively showcased production workflow and capabilities.

Eikon Therapeutics

Hayward, CA

Mechanical Engineering Intern

June 2024 - August 2024

- Increased laser precision by 44.4% and reduced alignment time by 90% by designing and implementing automated optical test rig using Python-controlled hardware, beam profiler, and custom GUI for non-programming engineers.
- Contributed to productization of advanced microscope by validating compact designs to enable transition from server rack prototype to deployable research tool; findings influenced testing methodology for ongoing development.

Design Visionaries

San Jose, CA

Design Intern

May 2023 - August 2023

- Filed 4 USPTO patents for small businesses by designing and prototyping innovative medical and consumer devices in Siemens NX; validated concepts through DFM analysis and 3D printing including volume-regulated drinking cups.
- Increased intern design productivity by offering comprehensive Siemens NX training, eliminating hand drafting.

Research and Club Experience

Musculoskeletal Biomechanics Lab

New York, NY

Undergraduate Researcher

September 2024 - Present

- Increased knee-implant testing throughput by converting scans to CAD, designing 3D-printed fixtures, and building LabVIEW data acquisition system with solenoid, LVDT, and load cells for NITRO living knee replacement validation.

Creative Machines Lab

New York, NY

Undergraduate Researcher

February 2024 - Present

- Delivered low-cost food 3D printer for Michelin-star chef by iterating CAD models in SolidWorks, generating optimized G-code in Python and rapid prototyping to improve print consistency and speed.

Columbia Makerspace

New York, NY

Superuser Volunteer

January 2024 - Present

- Hold office hours and support students with various academic and personal projects, regularly training users on and maintaining equipment (laser cutter, 3D printer, waterjet, woodshop, embroidery machine).

Engineering and Design Projects

Handheld Audiobook Player

- Building handheld audiobook player using Raspberry Pi and 3D-printed enclosure designed in Fusion360/Photoshop.

Pinball Machine

- Built an F1-themed pinball machine from scratch by reverse-engineering components at a local pinball museum, designing a custom playfield in Fusion360, and integrating CNC-machined parts with electronics for gameplay.

Multi Resistance Mechanism Foot Brace

- Led 6-person team to develop a medical device for foot drop patients by conducting user research with multiple sclerosis patients, iterating designs based on feedback, and prototyping a multi-resistance foot brace.

Skills & Interests

Skills: Siemens NX • Solidworks • Blender • LabVIEW • JMP • Python • C • Raspberry Pi • Adobe Suite • MS Office

Manufacturing: DFM/DFA • DOE • Injection Molding • 3D Printing • CNC Machining • Laser Cutting • Waterjet

Interests: Powerlifting • DJing • 3D Modeling • Reading • Board Games • Needlework • Book Binding • Making

Other apple things:

1. Made presentation boards to pitch proposals to ID such as various cost downs like removing PVD in favor for a natural stainless steel that can be color shifted using INCO and acid treatments during passivation process
2. Fixing corrosion failures by going in person to vendor sites and working on acid treatments and passivation processes
3. Leading regular vendor calls and cross function team syncs (Sensing, PD, REL, MD, HWE, color) to improve cosmetic and functional capability of user facing components.
4. Met with subject matter experts hellu to learn about improvements to passivation, plating, PVD, material surface science, adhesives, etc
5. Working with vendors to develop new DFMs to reduce assembly costs
6. 24% yield window cosmetics to 98.9% for one FAI (gap) and 44% to 100% for another (step)